

# Welfare, and Integrability

Advanced Microeconomics 1: Economic Decision Making. Lecture 4

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## The state of play

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The question: Who did inflation hurt, is wage indexation the right policy response?

1. **L1**: we can formally pose the question.
2. **L2**: the ideal answer assuming  $u$  is  $e(p^1, u^0) - w$ , per household.
3. **L3**: the building blocks for not assuming  $u$ .
4. **Today**: (1) putting together the pieces  $\rightarrow$  finding the answer from data, (2) interrogate the remaining assumptions.

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## **Recovering CV using demand functions**

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## The integrability problem

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Given data a demand function  $x(p, w)$  what can we learn about the underlying preference relation?

### Theorem · integrability (MWG 3.H)

If  $x(p, w)$  satisfies Walras' law, h.d.0, and has a symmetric, negative semidefinite Slutsky matrix, then there exists a rational preference relation rationalising  $x$ .

Proof → p. 25

Proof: sketch in the appendix; full treatment MWG 3.H.

## An alternative approach: Revealed Preferences

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### Definition · SARP (Houthakker 1950)

$x(p, w)$  satisfies **SARP** if for any chain  $(p^1, w^1), \dots, (p^N, w^N)$  with  $x(p^{n+1}, w^{n+1}) \neq x(p^n, w^n)$  and  $p^n \cdot x(p^{n+1}, w^{n+1}) \leq w^n$  for all  $n \leq N - 1$ , we have  $p^N \cdot x(p^1, w^1) > w^N$ .

### Theorem · Houthakker (1950)

$x(p, w)$  satisfies SARP  $\iff$  there is a rational preference relation that rationalizes  $x$ .

Proof deferred (Houthakker 1950; MWG 3.J).

One difference: SARP doesn't need the derivatives of  $x$  to exist.

## What does this theorem get us?

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1. Find a demand function  $x$  such that:

1.1 The following hold (1) h.d.0, (2) Walras, (3) symmetric NSD S

1.2 OR  $x(p, w)$  satisfies SARP

⇒ there exists a rational  $u$  that generated  $x$ .

⇒  $x$  is the product of optimisation.

⇒  $v, e, h$  exist and we can use Shephard, Slutsky, and Roy.

⇒ we can then recover  $v$  and thus  $e$ , and thus  $CV$ .

## In practice

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1. Check your demand function  $x(p, w)$  is h.d.0, satisfies Walras and its  $S$  is symmetric NSD.
2. Define  $w(p) = e(p, u^0)$  the wealth needed to maintain  $u^0$  as some  $p$  changes.
3. **Shephard** gets you  $h(p, u^0) = \frac{de(p, u^0)}{dp}$ .
4. Duality implies  $h(p, u^0) = x(p, e(p, u^0)) = x(p, w(p))$ .
5. Thus we have constructed the ODE:  $\frac{dw(p)}{dp} = x(p, w(p))$ .
6. Solve:  $-CV = e(p^1, u^0) - w = \int_{p_0}^{p^1} x(p, w(p)) dp$

## But we need a demand function

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- So we have our answer,  $CV$ , estimable without assuming a utility function  
$$-CV = \int_{p_0}^{p^1} x(p, w(p)) dp.$$
- But, we still need a demand function  $x(p, w)$ .
- We don't have to assume this, we can fit it to data.
- Enter econometrics!

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**Taking this to our question**

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## In practice, for Belgium 2022: the ingredients

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- **The shock:** as before home energy +57.9% over 2022 (NBB), so  $g = 0.579$ .
- **Demand function:** *Stone–Geary*,  $x_E(p, w) = \gamma + \frac{\beta}{p}(w - p\gamma)$ .
- **The elasticity:** residential energy demand is *price-inelastic*. Belgian causal estimate (electricity, VAT-reform quasi-experiment):  $\eta \in [-0.17, -0.09]$  (Hindriks and Serse 2022). We calibrate  $(\beta, \gamma)$  so the form's share and elasticity match  $(s_0, \eta)$  at  $p^0$ .

## In practice, for Belgium 2022: the answer

Solving  $\frac{dw}{dp} = \gamma(1 - \beta) + \frac{\beta}{p} w$  (linear; integrating factor  $p^{-\beta}$ ):

$$\frac{-CV}{w} = \frac{\gamma p^1 + (w - \gamma p^0) (p^1 / p^0)^\beta}{w} - 1$$

With  $g = 57.9\%$ , and  $\eta = -0.13$ , recovered  $-CV/w$ :

|                    | fixed basket<br>(Laspeyres) | data elasticity<br>(Stone–Geary, $\eta = -0.13$ ) | Cobb–Douglas<br>( $\gamma = 0$ ) |
|--------------------|-----------------------------|---------------------------------------------------|----------------------------------|
| poor, $s_0 = 0.10$ | 5.8%                        | <b>5.7%</b>                                       | 4.7%                             |
| rich, $s_0 = 0.06$ | 3.5%                        | <b>3.4%</b>                                       | 2.8%                             |

Proof → p. 27

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## Checking our assumptions

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## With a demand function $x(p, w)$ in hand

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$x$  is generated by some rational preference relation iff...

1.  $x$  is (i) h.d.0, (ii) Walras, (iii) symmetric NSD Slutsky matrix.
2.  $x$  satisfies SARP.

## Non-parametric test with just data $\{(p^t, x^t)\}_{t=1}^T$ , no estimated demand function

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Let  $R$  denote “revealed preferred” i.e.  $x^t R x^s$  means: I chose  $x^t$  even though  $x^s$  was (weakly) available i.e.  $p^t \cdot x^s \leq w^t$ .  $P$  denotes the strict version.

### Definition · GARP

No chain  $x^{t_1} R x^{t_2} R \dots R x^{t_n}$  with  $x^{t_n} P x^{t_1}$  exists.

### Theorem · Afriat (1967)

Data satisfy GARP  $\Leftrightarrow$  Data is rationalisable by a continuous, increasing, concave utility function.

Proof  $\rightarrow$  p. 26

## Rationality of our demand function

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Energy demand  $x_E = \gamma(1 - \beta) + \frac{\beta w}{p_E}$ ; other good  $x_O = \frac{(1-\beta)}{p_O}(w - p_E\gamma)$ . Write  $y = w - p_E\gamma$ .

1. **h.d.0.** Scale  $(p_E, p_O, w) \rightarrow (tp_E, tp_O, tw)$ , run-through,  $t$ 's cancel ✓

2. **Walras' law.**  $p_E x_E + p_O x_O = (p_E\gamma + \beta y) + (1 - \beta)y = p_E\gamma + y = w$ . ✓

3. **Slutsky symmetric & NSD.**  $S_{\ell k} = \frac{\partial x_\ell}{\partial p_k} + \frac{\partial x_\ell}{\partial w} x_k$ . Compute the cross terms:

$$S_{EO} = \frac{\beta(1 - \beta)y}{p_E p_O} = S_{OE}, \quad S_{EE} = -\frac{\beta(1 - \beta)y}{p_E^2} < 0, \quad S_{OO} = -\frac{\beta(1 - \beta)y}{p_O^2} < 0.$$

Own-price terms negative, and  $\det S = 0 \Rightarrow S$  is negative semidefinite. ✓

## Our assumptions

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1. **Rationality** — choices reveal a stable ranking.
2. **No money illusion** — h.d.0.
3. **Stable preferences** — the  $u^0$  being restored is the same object across the comparison.
4. **A choice of reference** — CV or EV; we treated it as a legitimate choice.

## Assumption 1: rationality — the evidence

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Does the observable footprint (GARP) actually hold when we look?

- **For:** in the largest test — ~2,000 Dutch adults, portfolio choices — consistency far exceeds random-choice benchmarks (Choi, Kariv, Müller & Silverman, AER 2014). Rationality is not universal, but it is the modal behaviour.
- **Against:** consistency varies sharply across people — lower for the poor, less educated, older, and those under cognitive load or stress (Kim et al., *Science* 2018; CKMS 2014). Never holds exactly or always.

## Assumption 2: no money illusion — the evidence

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h.d.0 says scaling all prices and wealth changes nothing.

- **For:** over long horizons and with salient information, real variables always dominate.
- **Against:** in the short run, no. Nominal framing systematically moves choices — Shafir, Diamond & Tversky (QJE 1997) show people reason in nominal terms about wages, contracts and prices even when told the inflation rate. A 2% wage raise at 4% inflation *feels* like a gain.
- **Why it matters here:** indexing exists because people worry about the “real”.

### Assumption 3: stable preferences — the evidence

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The  $u^0$  we restore must be the *same* object before and after the shock. Is it?

- **Against:** preferences are often *constructed at the moment of choice*, not retrieved from a fixed ranking. Normatively-equivalent elicitations give systematically different answers — **preference reversals**: people choose  $A$  over  $B$  yet price  $B$  above  $A$  (Lichtenstein & Slovic 1971, replicated on a Las Vegas casino floor; survives incentives, Grether & Plott 1979). If the ranking depends on how you ask, “the same  $u^0$ ” is not innocent.
- **For (the defence):** reversals are largest in unfamiliar, one-shot lab tasks. For *repeated, familiar* decisions — groceries, energy, the household budget — choices are far more stable, which is exactly the domain we care about.
- It seems like stability is a reasonable *working* assumption for routine consumption, and a fragile one for novel or framed choices.

## Assumption 4: the reference point — the evidence

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CV and EV have different reference points, this shouldn't matter...

- **Against:** people value losses from a reference far more than equivalent gains — **loss aversion** (Kahneman & Tversky 1979). So the *endowment effect*: willingness-to-accept to give up a good routinely exceeds willingness-to-pay to acquire it, by a factor of  $\sim 2$  (Kahneman, Knetsch & Thaler, JEP 1991).
- **Why it matters here:** which reference the policy adopts — the old standard of living ( $u^0$ , CV) or the new ( $u^1$ , EV) — now matters.

### Behavioural Economics

Kahneman, Knetsch & Thaler (JEP 1991): “The Endowment Effect, Loss Aversion, and Status Quo Bias”, enter **behavioural economics**.

## Where does that leave us

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1. Rationality. **OK on average** — but heterogenous along income, education, stress, etc.
2. No money illusion. **Holds in the long-run.**
3. Stable preferences. **Holds more for regular than rare purchases**, verdict still fairly open.
4. Reference point. **Dead.** Certainly matters, how much is a little unclear.

We can extend the framework to allow for deviations to each assumption — for another day :)

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## Summing up

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### Takeaways

- **(a) Who did the 2022 energy shock hurt, and by how much?** Everyone by 3-6%, the relatively poorer by more.
- **(b) Is one-basket indexation the right compensation?** No — it over-pays the average and misses both tails.
- **Next:** Firms, where do those price rises come from?

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## **Appendix: proofs**

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## Proof sketch: integrability (MWG 3.H)

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**Step 1 — recover  $e$  from  $x$ .** Fix  $(p^0, w^0)$  and try to solve the system of PDEs given by Shephard run backwards:

$$\nabla_p e(p) = x(p, e(p)), \quad e(p^0) = w^0.$$

A solution exists locally iff the system is consistent — Frobenius' condition: the “cross-derivatives” of the right-hand side must commute. Computing them, the condition is exactly **symmetry of  $S(p, e(p))$** . That is the moment symmetry stops being decoration: it is the solvability condition for reconstructing the cost of living from demand.

**Step 2 — recover preferences from  $e$ .** Given  $e(\cdot, u)$  (concave, by NSD, in the recovered system), define upper contour sets as the intersections of half-spaces

$$V_u = \{x \in \mathbb{R}_+^L : p \cdot x \geq e(p, u) \text{ for all } p \gg 0\}.$$

Concave-duality (support-function) arguments show the preference relation these sets induce is rational and generates  $x$ . □ (sketch; full argument MWG 3.H.1–3.H.2)

## Proof sketch: Afriat's theorem

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**(iii)  $\Rightarrow$  (i):** a rationalizing utility orders the data; revealed-preference chains inherit the order, so no cycle can end strictly above its start.

**(i)  $\Rightarrow$  (ii):** GARP makes the revealed relation acyclic, so observations can be ordered; a combinatorial induction then picks  $U_t, \lambda_t > 0$  satisfying the inequalities — read as “a concave function with slope  $\lambda_t p^t$  at  $x^t$  leaves room for  $U_s$  at  $x^s$ .”

**(ii)  $\Rightarrow$  (iii):** build the utility explicitly as the lower envelope of those supporting hyperplanes:

$$u(x) = \min_t \{ U_t + \lambda_t p^t \cdot (x - x^t) \}$$

— continuous, increasing, concave by construction, and it rationalizes every observation.



Sketch; Afriat (1967), Varian (1982).

## Derivation: solving the Stone–Geary recovery ODE

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Stone–Geary energy demand  $x_E(p, w) = \gamma + \frac{\beta}{p}(w - p\gamma) = \gamma(1 - \beta) + \frac{\beta}{p} w$ , so the recovery ODE  $\frac{dw}{dp} = x_E(p, w(p))$  is

$$\frac{dw}{dp} - \frac{\beta}{p} w = \gamma(1 - \beta).$$

**Linear, first order** — integrating factor  $\mu(p) = \exp(-\int \frac{\beta}{p} dp) = p^{-\beta}$ . Multiply through:

$$\frac{d}{dp} (w p^{-\beta}) = \gamma(1 - \beta) p^{-\beta} \implies w p^{-\beta} = \gamma p^{1-\beta} + C.$$

So  $w(p) = \gamma p + C p^{\beta}$ . **Pin  $C$  from the start:**  $w(p^0) = w$  gives  $C = (w - \gamma p^0) (p^0)^{-\beta}$ , hence

$$w(p) = \gamma p + (w - \gamma p^0) \left( \frac{p}{p^0} \right)^{\beta}.$$

Evaluate at  $p^1$ :  $-CV = w(p^1) - w$ , and  $\frac{-CV}{w} = \frac{\gamma p^1 + (w - \gamma p^0)(p^1/p^0)^{\beta} - w}{w} - 1$ .

## Derivation: calibrating $(\beta, \gamma)$ to the data

Normalise  $p^0 = 1, w = 1$ ). Targetes: the energy share  $s_0$  and the own-price elasticity  $\eta$ . Stone–Geary energy demand  $x_E = \gamma(1 - \beta) + \frac{\beta w}{p}$ , so at the initial point  $x_E^0 = \gamma + \beta(1 - \gamma)$ .

**Target 1 — share.**  $s_0 = \frac{p x_E}{w} = x_E^0 = \gamma + \beta(1 - \gamma)$ .

**Target 2 — elasticity.**  $\frac{\partial x_E}{\partial p} = -\frac{\beta w}{p^2}$ , so  $\eta = \frac{\partial x_E}{\partial p} \cdot \frac{p}{x_E} = -\frac{\beta}{x_E^0} = -\frac{\beta}{s_0}$ .

Target 2 gives  $\beta$  directly; substitute into Target 1 for  $\gamma$ :

$$\beta = -\eta s_0, \quad \gamma = \frac{s_0(1 + \eta)}{1 + \eta s_0}.$$

- Poor ( $s_0 = 0.10, \eta = -0.13$ ):  $\beta = 0.013, \gamma = 0.088$ . So  $\gamma/s_0 \approx 88\%$  of energy is *subsistence* — the can't-cut-it floor. Only  $\sim 12\%$  is price-responsive: that *is* the inelasticity.